GURU99

How to Install Hadoop with Step by Step Configuration on Linux Ubuntu

By :

Prafulla Ranadive

Hours Updated June 13, 2024

In this tutorial, we will take you through step by step process to install Apache Hadoop on a Linux box (Ubuntu). This is 2 part process

- Part 1) Download and Install Hadoop
- Part 2) Configure Hadoop

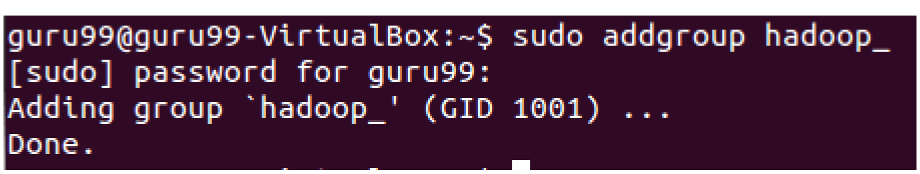
There are 2 **Prerequisites**

- You must have Ubuntu installed and running
- You must have Java Installed.

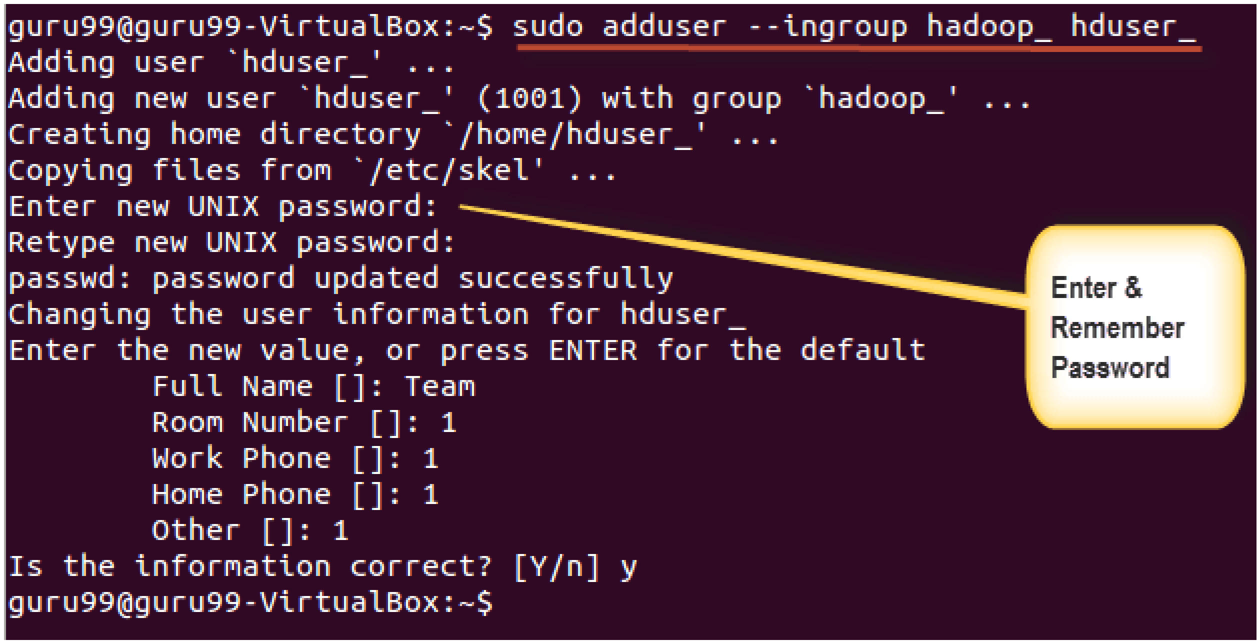
Part 1) Download and Install Hadoop

Step 1) Add a Hadoop system user using below command

```
sudo addgroup hadoop_
```




```
sudo adduser --ingroup hadoop_ hduser_
```

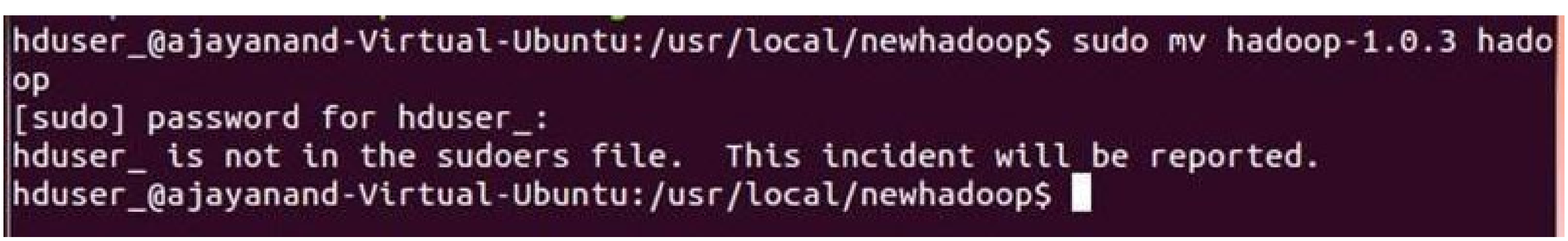


```
guru99@guru99-VirtualBox:~$ sudo adduser --ingroup hadoop_ hduser_
Adding user `hduser_' ...
Adding new user `hduser_' (1001) with group `hadoop_' ...
Creating home directory `/home/hduser_' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for hduser_
Enter the new value, or press ENTER for the default
        Full Name []: Team
        Room Number []: 1
        Work Phone []: 1
        Home Phone []: 1
        Other []: 1
Is the information correct? [Y/n] y
guru99@guru99-VirtualBox:~$
```


Enter your password, name and other details.

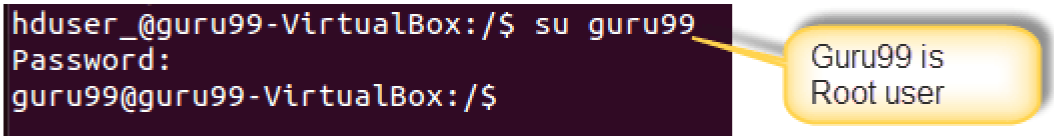
NOTE: There is a possibility of below-mentioned error in this setup and installation process.

"hduser is not in the sudoers file. This incident will be reported."



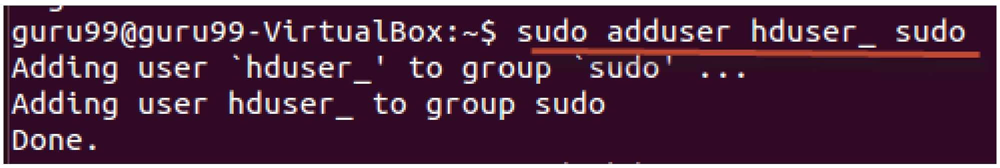
```
hduser_@ajayanand-Virtual-Ubuntu:/usr/local/newhadoop$ sudo mv hadoop-1.0.3 hado
op
[sudo] password for hduser_:
hduser_ is not in the sudoers file.  This incident will be reported.
hduser_@ajayanand-Virtual-Ubuntu:/usr/local/newhadoop$
```

This error can be resolved by Login as a root user

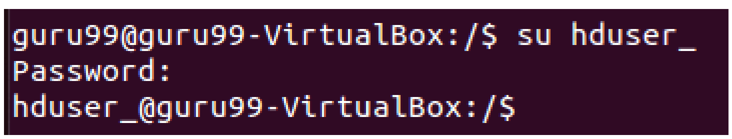


Execute the command

```
sudo adduser hduser_ sudo
```




```
Re-login as hduser_
```

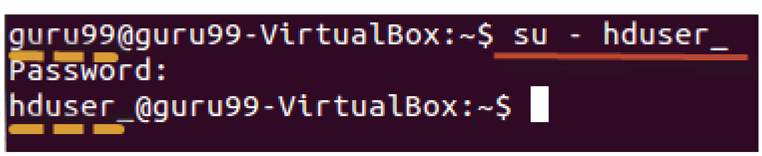


Step 2) Configure SSH

In order to manage nodes in a cluster, Hadoop requires SSH access

First, switch user, enter the following command

```
su - hduser_
```

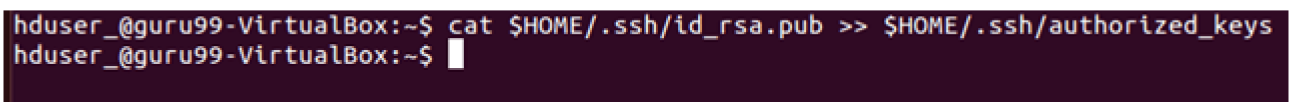


This command will create a new key.

```
ssh-keygen -t rsa -P ""
```

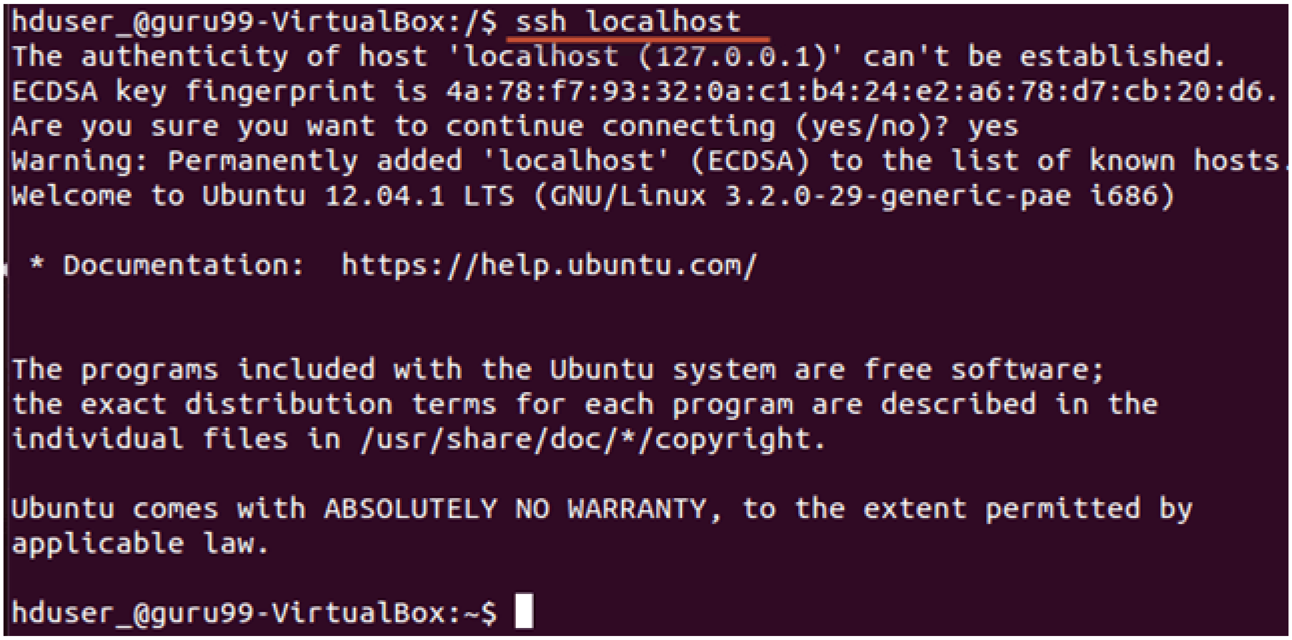
Enable SSH access to local machine using this key.

```
cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
```



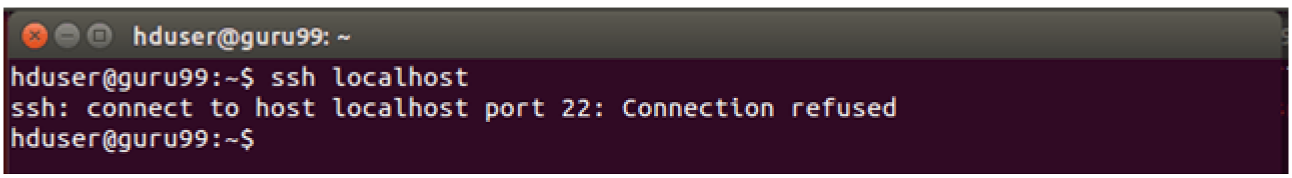
Now test SSH setup by connecting to localhost as 'hduser' user.

```
ssh localhost
```



```
hduser_@guru99-VirtualBox:/$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ECDSA key fingerprint is 4a:78:f7:93:32:0a:c1:b4:24:e2:a6:78:d7:cb:20:d6.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'localhost' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 12.04.1 LTS (GNU/Linux 3.2.0-29-generic-pae i686)

 * Documentation:  https://help.ubuntu.com/


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

hduser_@guru99-VirtualBox:~$
```

Note: Please note, if you see below error in response to 'ssh localhost', then there is a possibility that SSH is not available on this system-



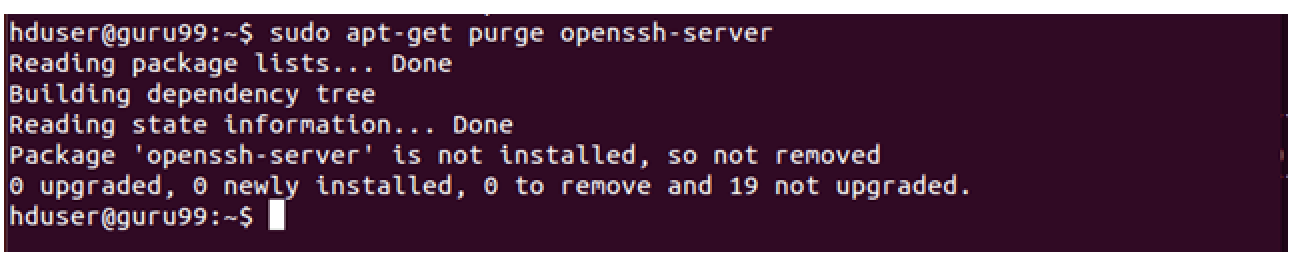
```
hduser@guru99: ~
hduser@guru99:~$ ssh localhost
ssh: connect to host localhost port 22: Connection refused
hduser@guru99:~$
```

To resolve this –

Purge SSH using,

```
sudo apt-get purge openssh-server
```

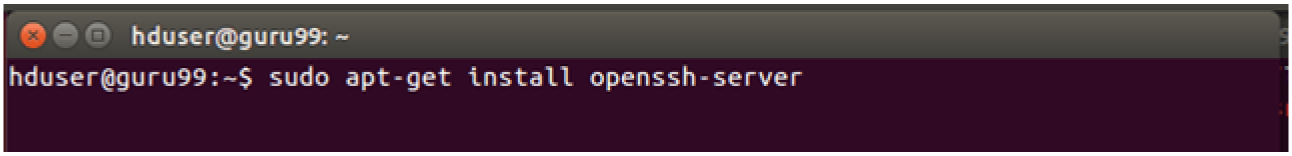
It is good practice to purge before the start of installation



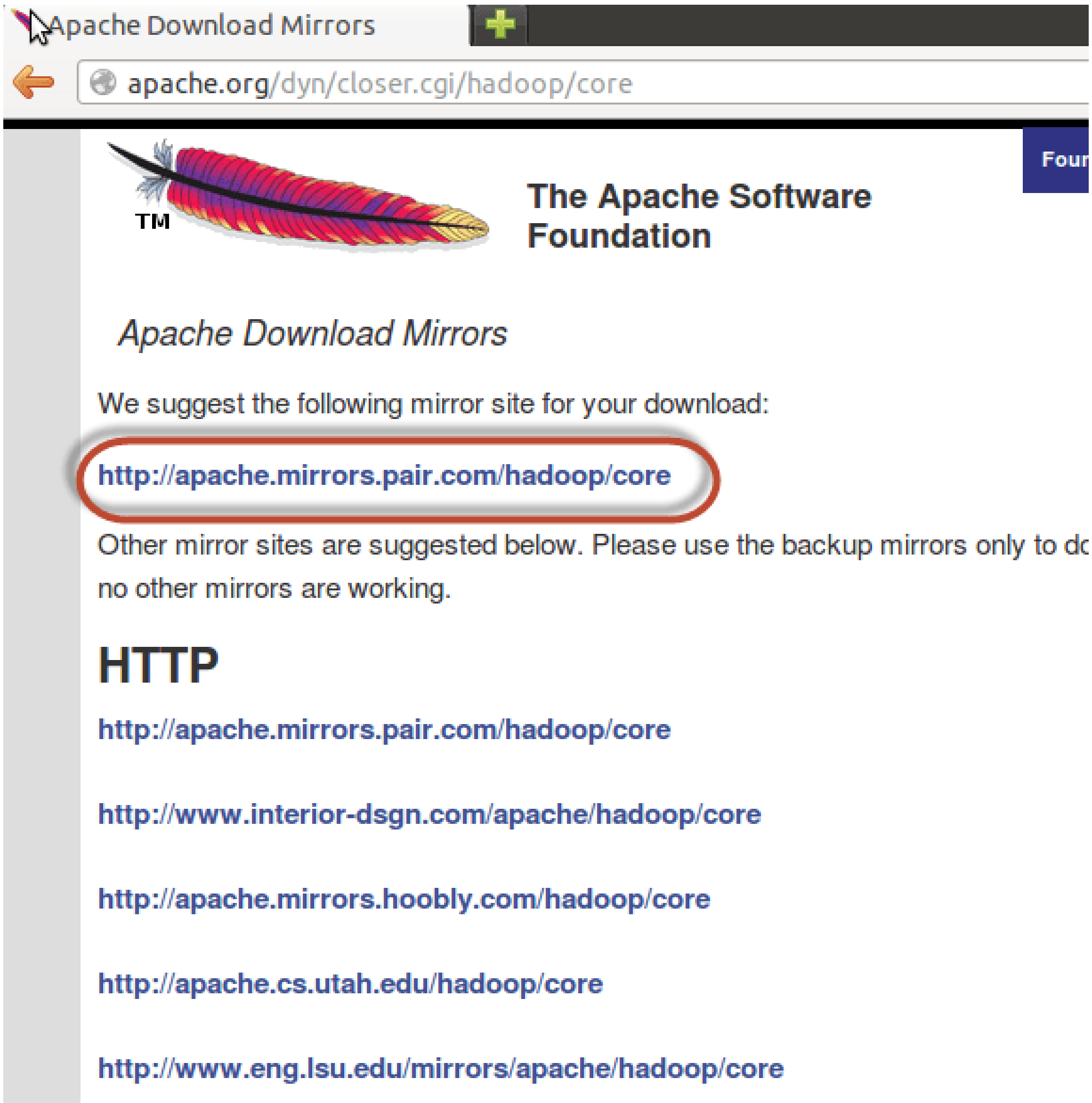
```
hduser@guru99:~$ sudo apt-get purge openssh-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
Package 'openssh-server' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
hduser@guru99:~$
```

Install SSH using the command-

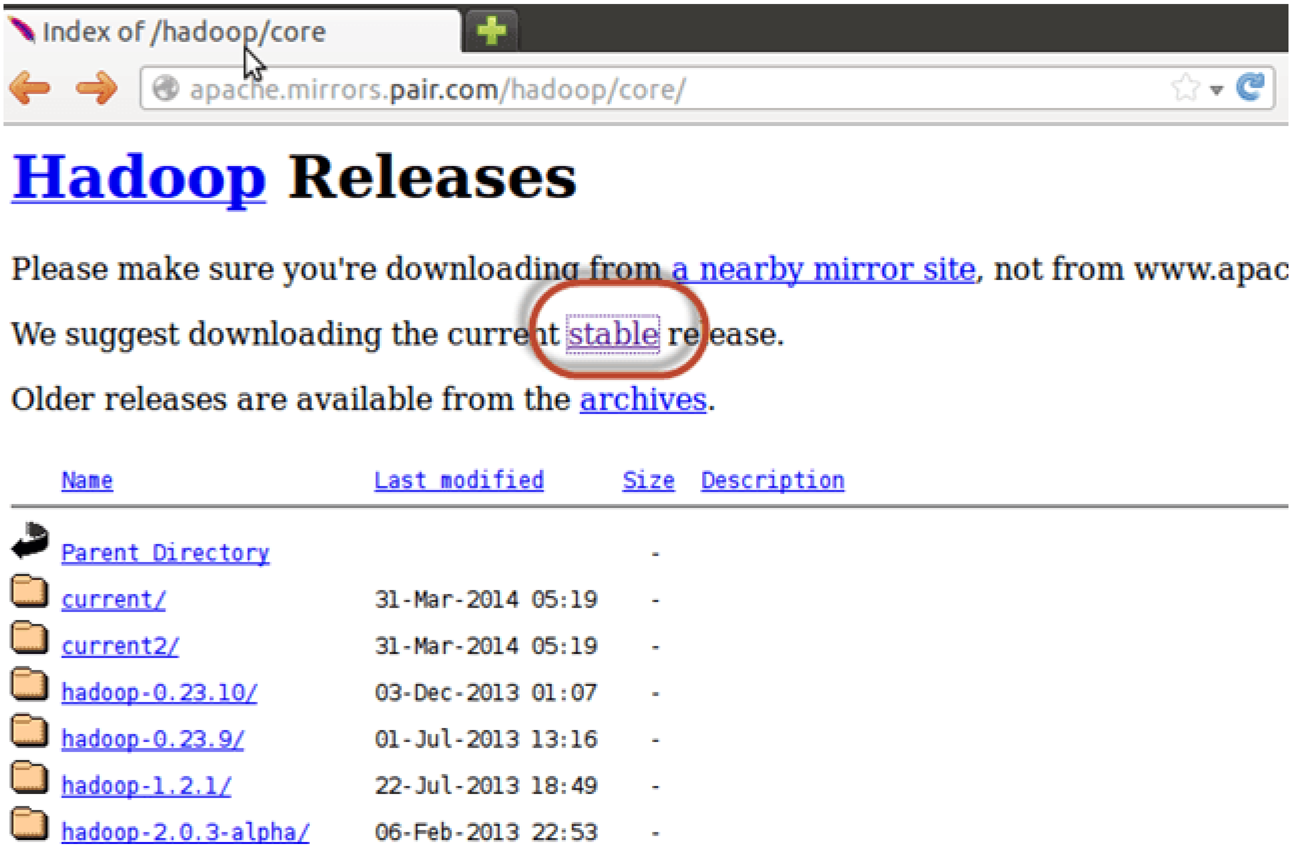
```
sudo apt-get install openssh-server
```



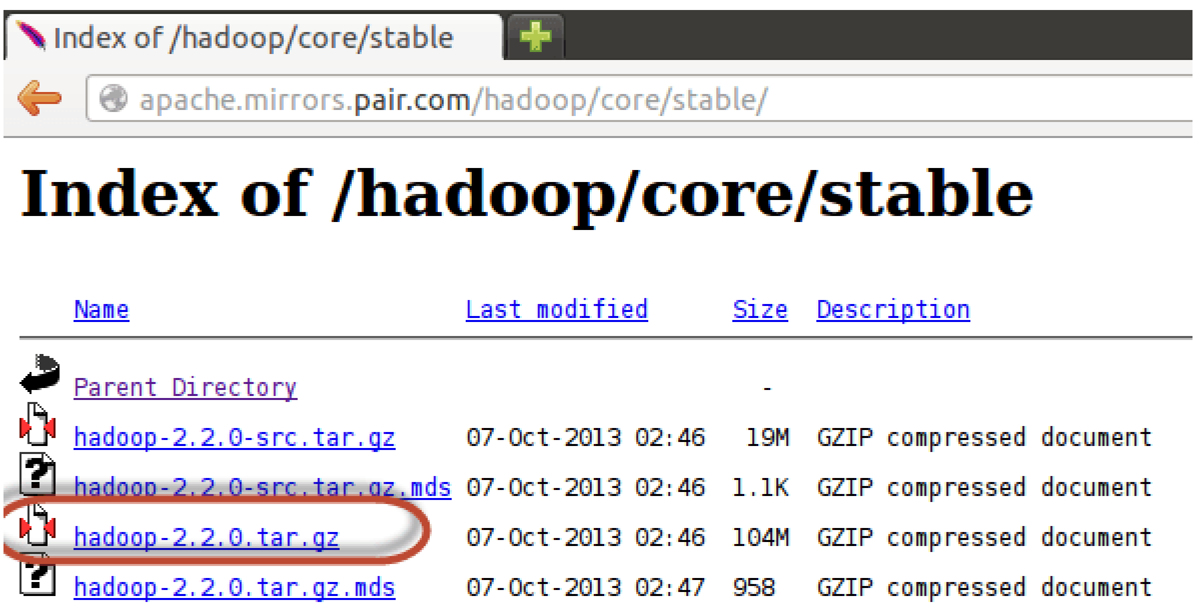
Step 3) Next step is to Download Hadoop



Select Stable



Select the tar.gz file (not the file with src)



Once a download is complete, navigate to the directory containing the tar file

```
hduser_@guru99-VirtualBox:~$ cd /home/guru99/Downloads
```

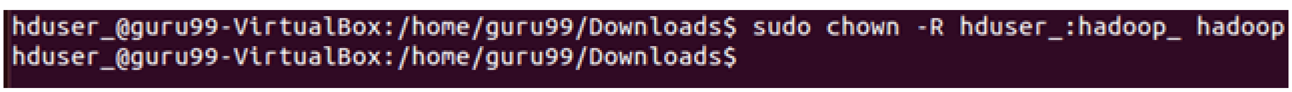
Enter,

```
sudo tar xzf hadoop-2.2.0.tar.gz
```

Now, rename hadoop-2.2.0 as hadoop

```
sudo mv hadoop-2.2.0 hadoop
```


```
sudo chown -R hduser_:hadoop_ hadoop
```



RELATED ARTICLES

→ What is Hadoop? Introduction, Architecture, Ecosystem, Components

→ HDFS Tutorial: Architecture, Read & Write Operation using Java API

→ Hadoop Pig Tutorial: What is Apache Pig? Architecture, Example

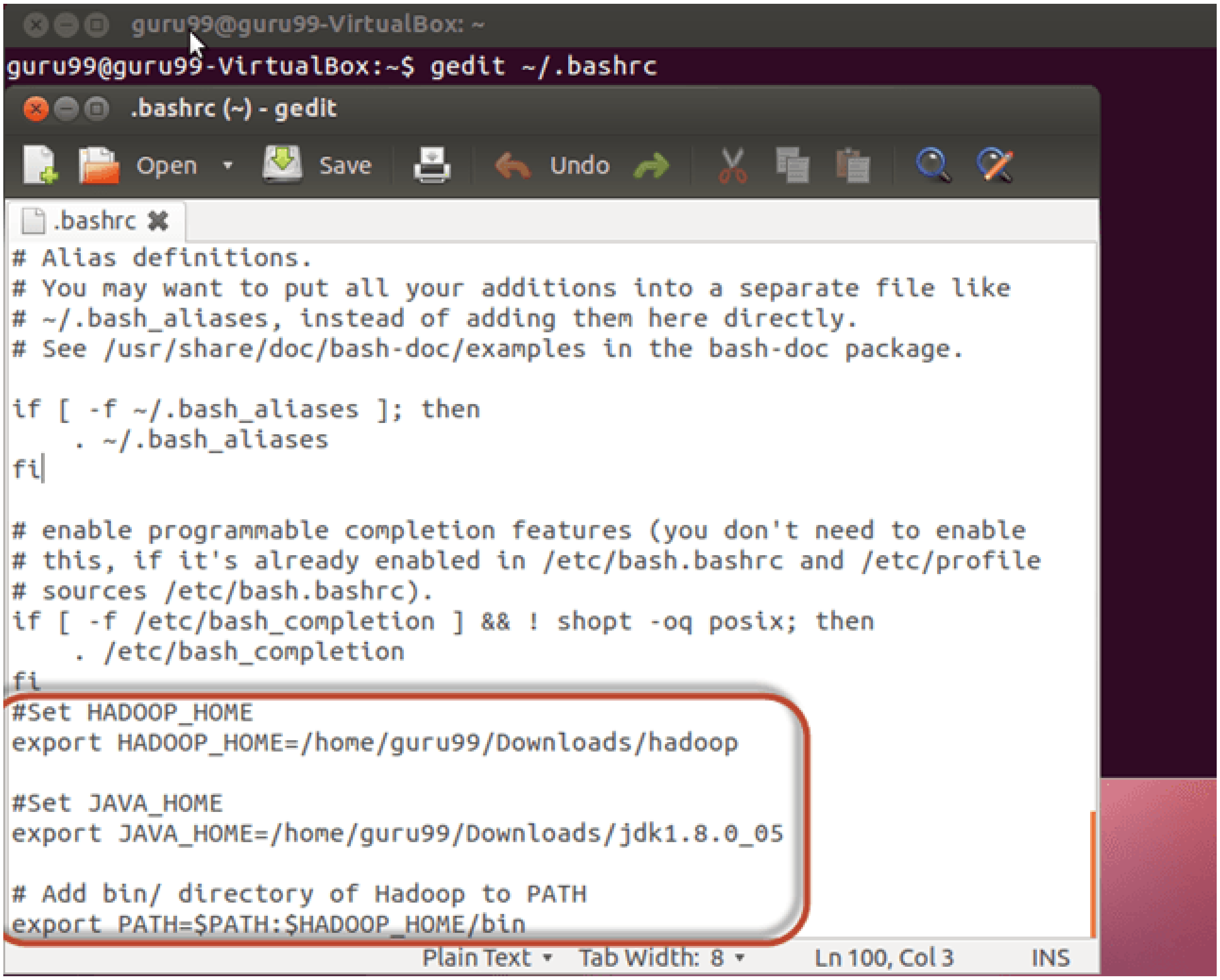
→ Talend Tutorial – What is Talend ETL Tool?

Part 2) Configure Hadoop

Step 1) Modify **~/.bashrc** file

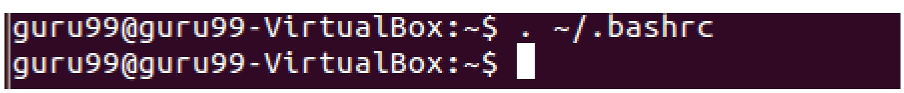
Add following lines to end of file **~/.bashrc**

```
#Set HADOOP_HOME
export HADOOP_HOME=<Installation Directory of Hadoop>
#Set JAVA_HOME
export JAVA_HOME=<Installation Directory of Java>
# Add bin/ directory of Hadoop to PATH
export PATH=$PATH:$HADOOP_HOME/bin
```



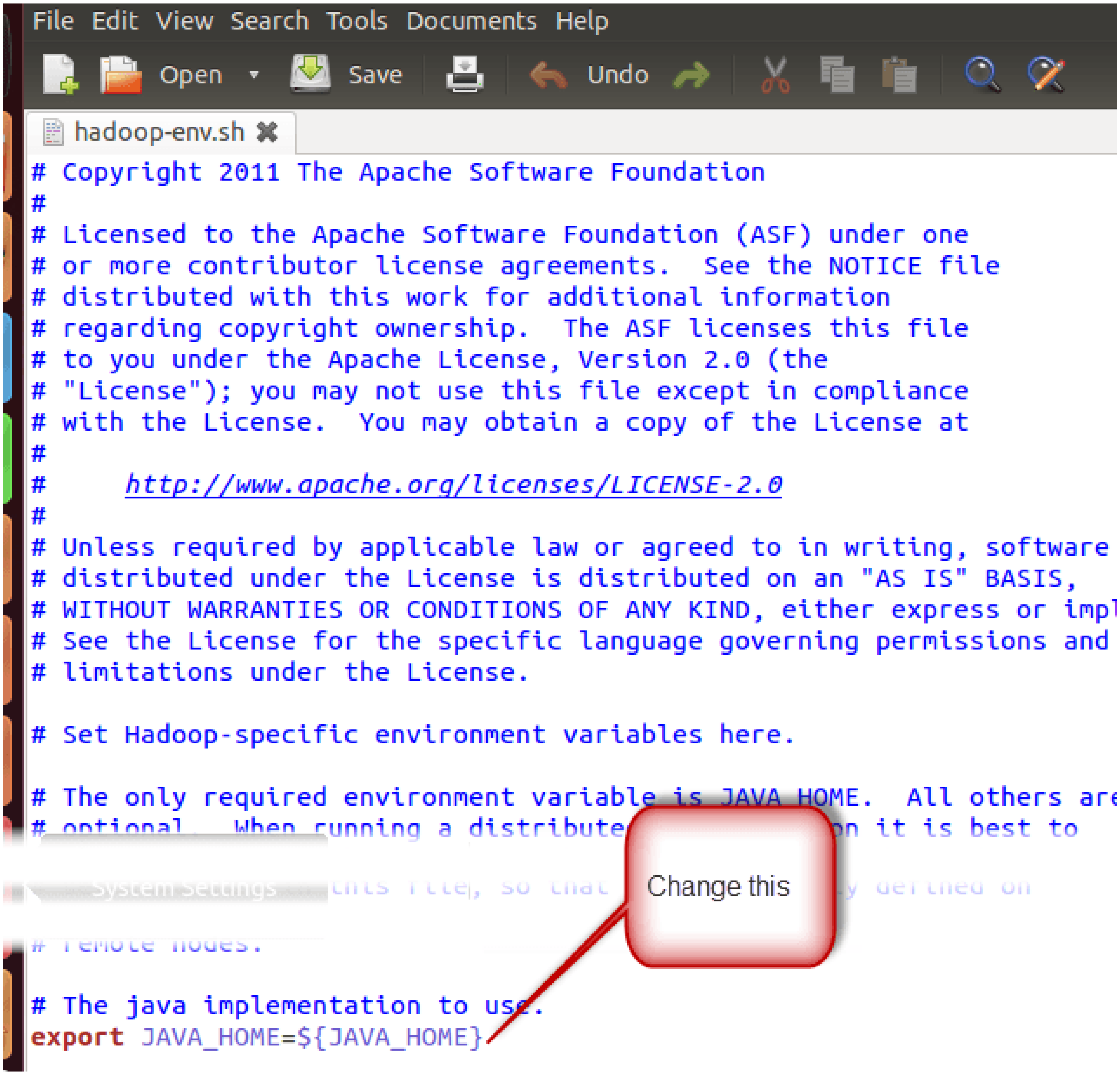
Now, source this environment configuration using below command

```
. ~/.bashrc
```



Step 2) Configurations related to HDFS

Set **JAVA_HOME** inside file **$HADOOP_HOME/etc/hadoop/hadoop-env.sh**


File Edit View Search Tools Documents Help
Open
Save
Undo
hadoop-env.sh
Copyright 2011 The Apache Software Foundation
#
Licensed to the Apache Software Foundation (ASF) under one
or more contributor license agreements. See the NOTICE file
distributed with this work for additional information
regarding copyright ownership. The ASF licenses this file
to you under the Apache License, Version 2.0 (the
"License"); you may not use this file except in compliance
with the License. You may obtain a copy of the License at
#
http://www.apache.org/licenses/LICENSE-2.0
#
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or impl
See the License for the specific language governing permissions and
limitations under the License.
Set Hadoop-specific environment variables here.
The only required environment variable is JAVA_HOME. All others are
The java implementation to use.
export JAVA_HOME=${JAVA_HOME}
Change this

With

```
# The java implementation to use.
export JAVA_HOME=/home/guru99/Downloads/jdk1.8.0_05
```

There are two parameters in **$HADOOP_HOME/etc/hadoop/core-site.xml** which need to be set-

1. 'hadoop.tmp.dir' – Used to specify a directory which will be used by Hadoop to store its data files.

2. 'fs.default.name' – This specifies the default file system.

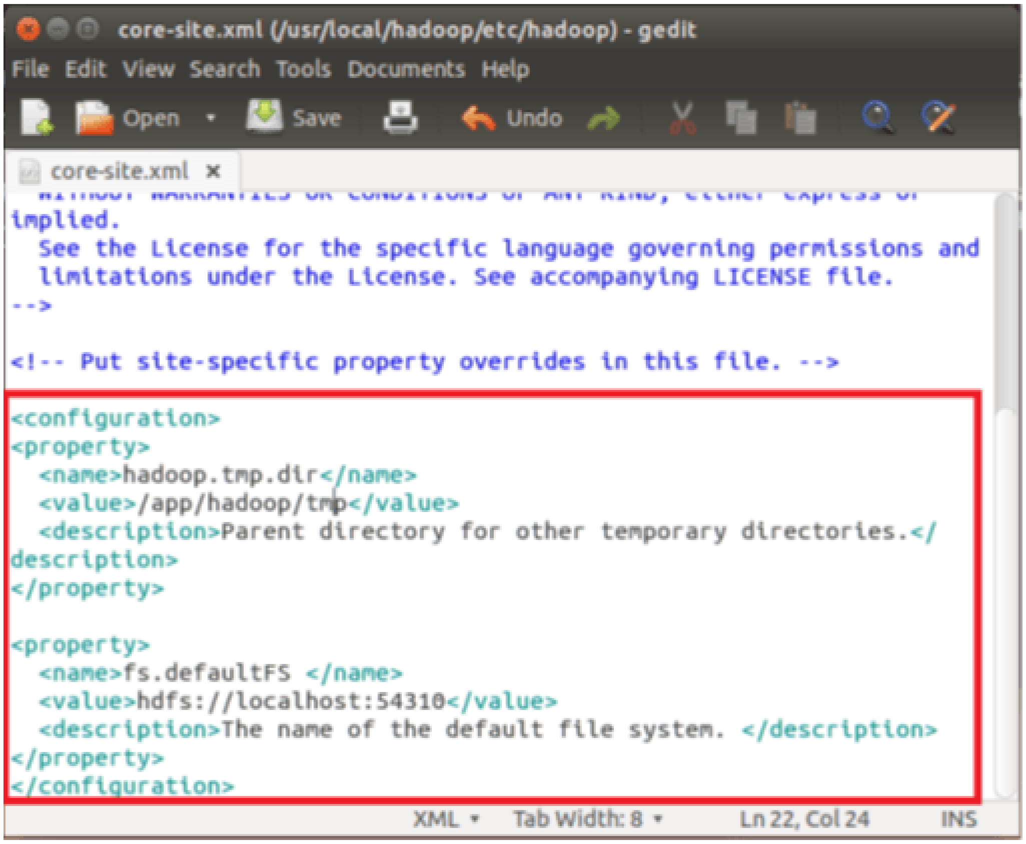
To set these parameters, open core-site.xml

```
sudo gedit $HADOOP_HOME/etc/hadoop/core-site.xml
```

```
guru99@guru99-VirtualBox:~$ sudo gedit /home/guru99/Downloads/hadoop/etc/hadoop/core-site.xml
```

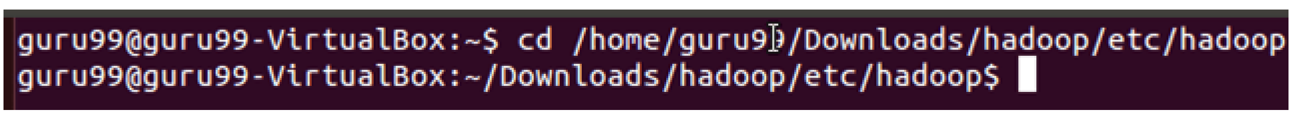
Copy below line in between tags <configuration></configuration>

```
<property>
<name>hadoop.tmp.dir</name>
<value>/app/hadoop/tmp</value>
<description>Parent directory for other temporary directories.</description>
</property>
<property>
<name>fs.defaultFS </name>
<value>hdfs://localhost:54310</value>
<description>The name of the default file system. </description>
</property>
```



```
core-site.xml (/usr/local/hadoop/etc/hadoop) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
core-site.xml
implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>hadoop.tmp.dir</name>
  <value>/app/hadoop/tmp</value>
  <description>Parent directory for other temporary directories.</
description>
</property>

<property>
  <name>fs.defaultFS </name>
  <value>hdfs://localhost:54310</value>
  <description>The name of the default file system. </description>
</property>
</configuration>
XML  Tab Width: 8  Ln 22, Col 24  INS
```

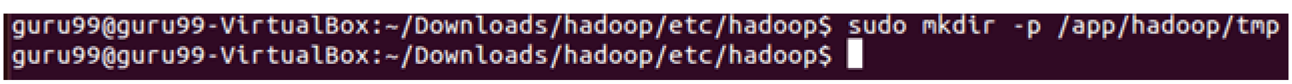
Navigate to the directory **$HADOOP_HOME/etc/Hadoop**



```
guru99@guru99-VirtualBox:~$ cd /home/guru99/Downloads/hadoop/etc/hadoop
guru99@guru99-VirtualBox:~/Downloads/hadoop/etc/hadoop$
```

Now, create the directory mentioned in core-site.xml

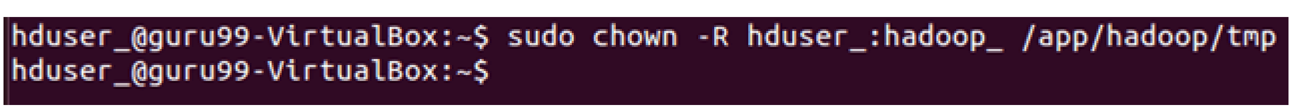
```
sudo mkdir -p <Path of Directory used in above setting>
```



```
guru99@guru99-VirtualBox:~/Downloads/hadoop/etc/hadoop$ sudo mkdir -p /app/hadoop/tmp
guru99@guru99-VirtualBox:~/Downloads/hadoop/etc/hadoop$
```

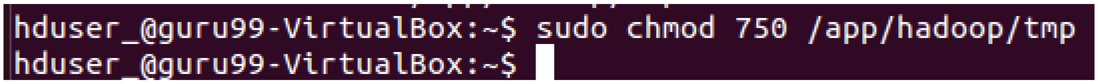
Grant permissions to the directory

```
sudo chown -R hduser_:Hadoop_ <Path of Directory created in above step>
```



```
hduser_@guru99-VirtualBox:~$ sudo chown -R hduser_:hadoop_ /app/hadoop/tmp
hduser_@guru99-VirtualBox:~$
```

```
sudo chmod 750 <Path of Directory created in above step>
```



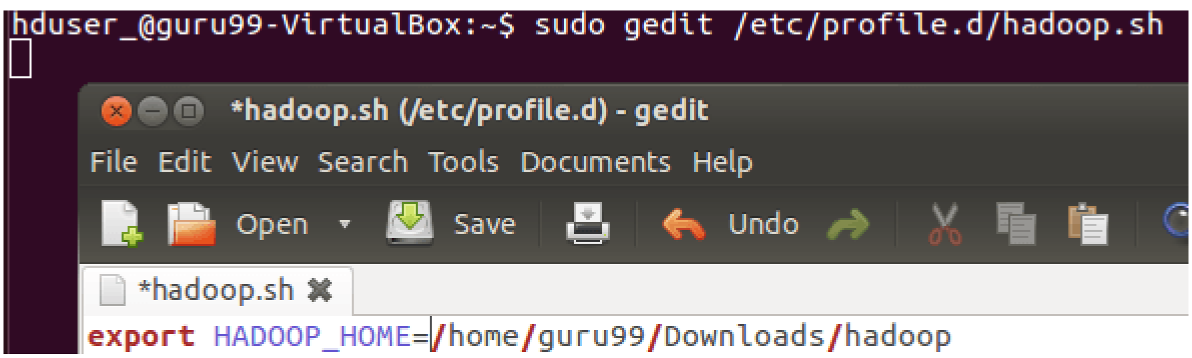
Step 3) Map Reduce Configuration

Before you begin with these configurations, lets set HADOOP_HOME path

```
sudo gedit /etc/profile.d/hadoop.sh
```

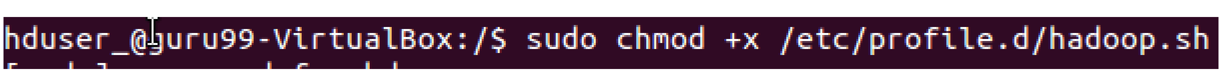
And Enter

```
export HADOOP_HOME=/home/guru99/Downloads/Hadoop
```



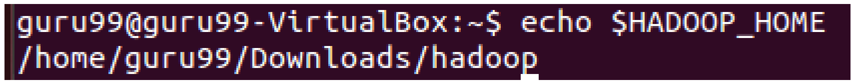
Next enter

```
sudo chmod +x /etc/profile.d/hadoop.sh
```



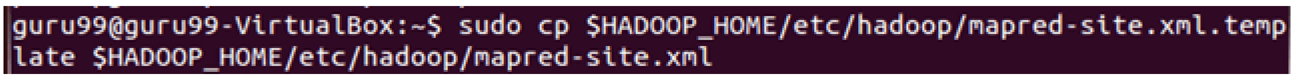
Exit the Terminal and restart again

Type echo $HADOOP_HOME. To verify the path



Now copy files

```
sudo cp $HADOOP_HOME/etc/hadoop/mapred-site.xml.template
$HADOOP_HOME/etc/hadoop/mapred-site.xml
```

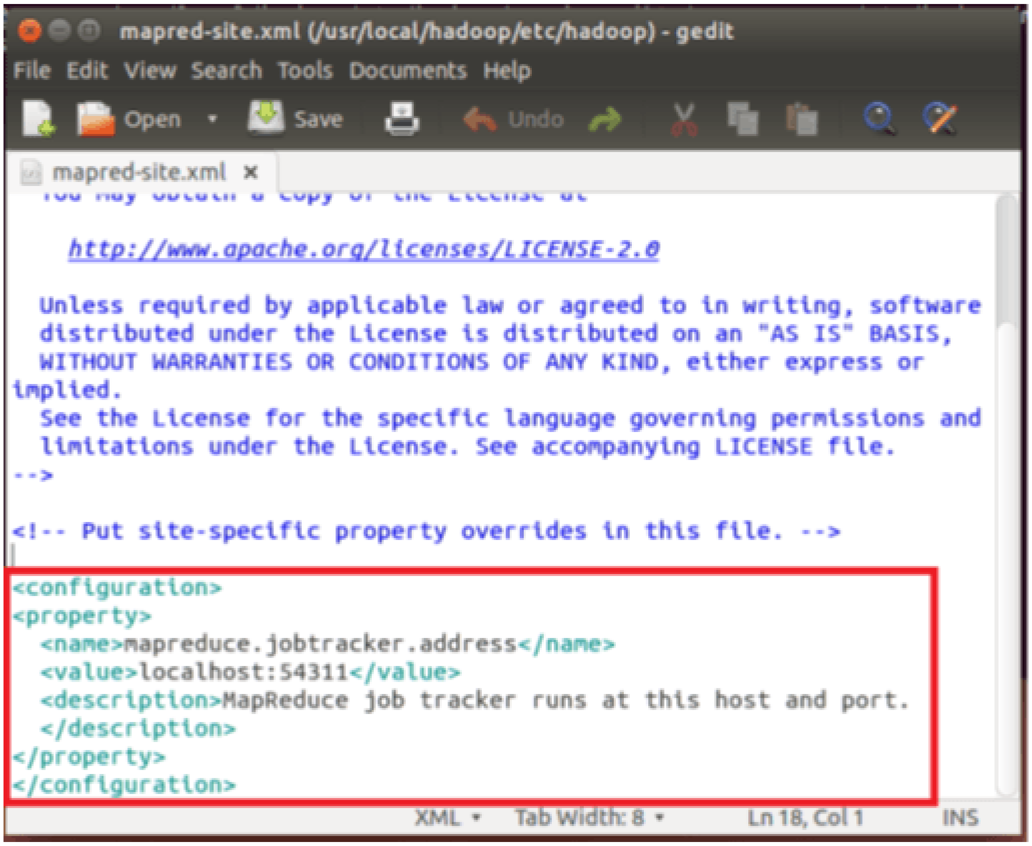


Open the **mapred-site.xml** file

```
sudo gedit $HADOOP_HOME/etc/hadoop/mapred-site.xml
```

Add below lines of setting in between tags <configuration> and </configuration>

```
<property>
<name>mapreduce.jobtracker.address</name>
<value>localhost:54311</value>
<description>MapReduce job tracker runs at this host and port.
</description>
</property>
```

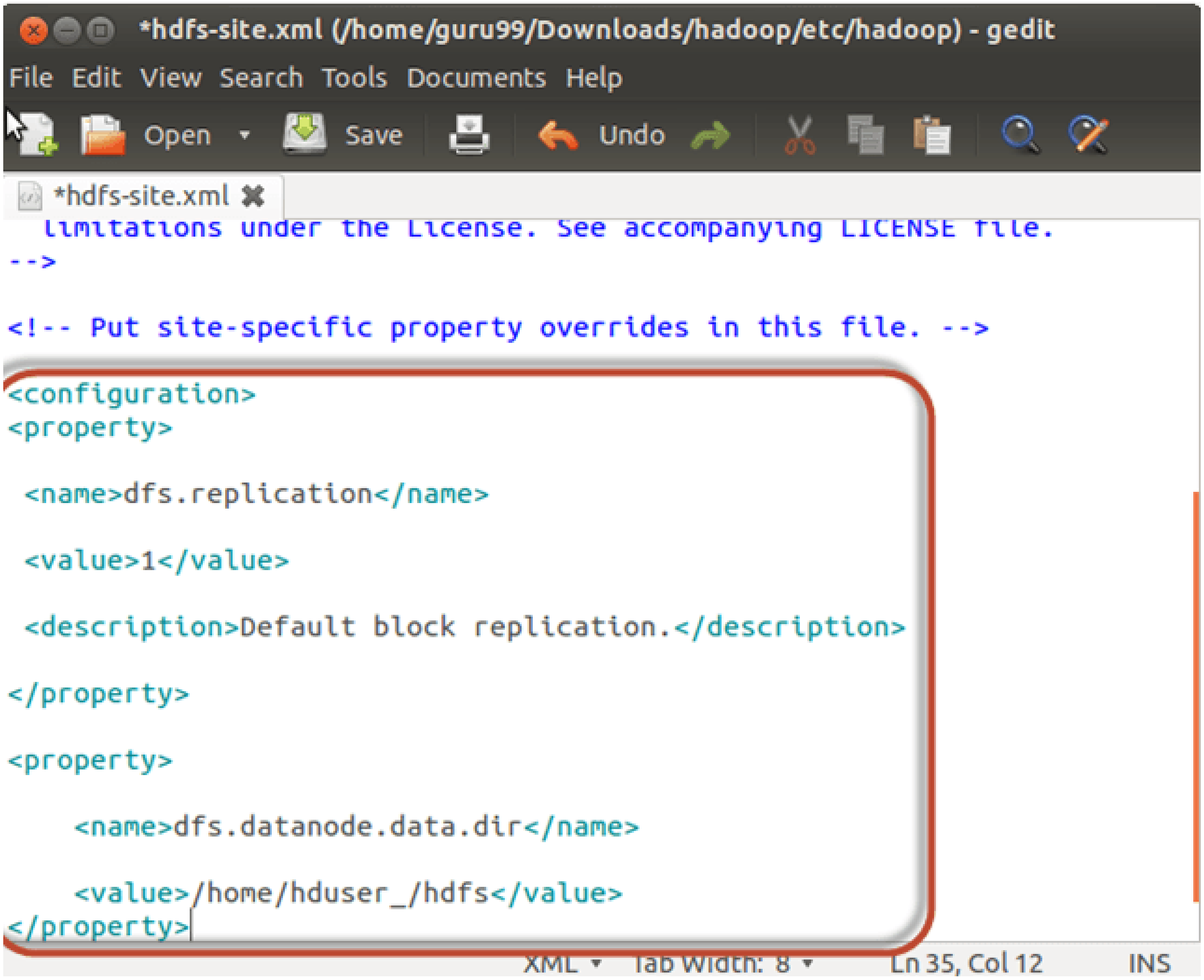


Open **$HADOOP_HOME/etc/hadoop/hdfs-site.xml** as below,

```
sudo gedit $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

Add below lines of setting between tags <configuration> and </configuration>

```
<property>
<name>dfs.replication</name>
<value>1</value>
<description>Default block replication.</description>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/home/hduser_/hdfs</value>
</property>
```



Create a directory specified in above setting-

```
sudo mkdir -p <Path of Directory used in above setting>
```

```
sudo mkdir -p /home/hduser_/hdfs
```


```
sudo chown -R hduser_:hadoop_ <Path of Directory created in above step>
```

```
sudo chown -R hduser_:hadoop_ /home/hduser_/hdfs
```

```
hduser_@guru99-VirtualBox:~$ sudo chown -R hduser_:hadoop_ /home/hduser_/hdfs
```

```
sudo chmod 750 <Path of Directory created in above step>
```

```
sudo chmod 750 /home/hduser_/hdfs
```

```
hduser_@guru99-VirtualBox:~$ sudo chmod 750 /home/hduser_/hdfs
```

Step 4) Before we start Hadoop for the first time, format HDFS using below command

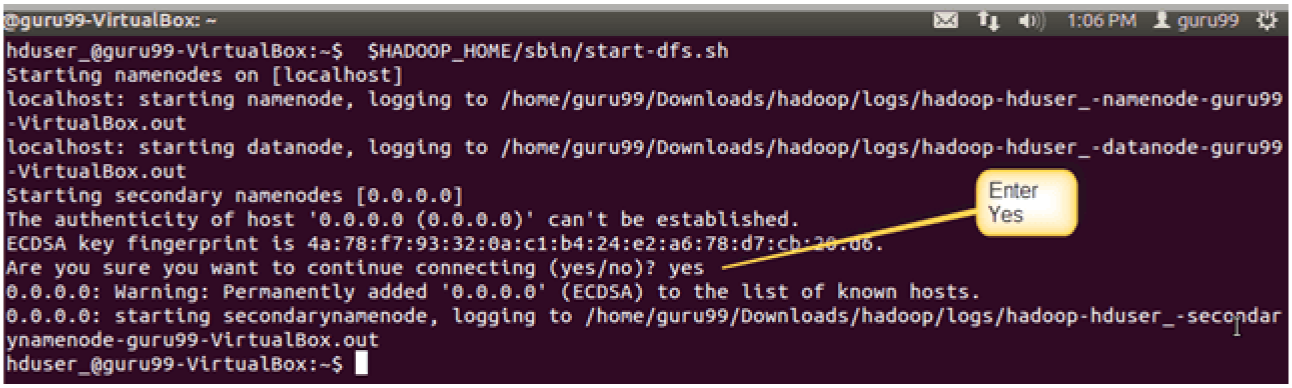
```
$HADOOP_HOME/bin/hdfs namenode -format
```

```
hduser_@guru99-VirtualBox:~$  $HADOOP_HOME/bin/hdfs namenode -format
14/05/05 13:01:58 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = guru99-VirtualBox/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 2.2.0
STARTUP_MSG:   classpath = /home/guru99/Downloads/hadoop/etc/hadoop:/home/guru99
/Downloads/hadoop/share/hadoop/common/lib/activation-1.1.jar:/home/guru99/Downlo
ads/hadoop/share/hadoop/common/lib/netty-3.6.2.Final.jar:/home/guru99/Downloads/
hadoop/share/hadoop/common/lib/protobuf-java-2.5.0.jar:/home/guru99/Downloads/ha
doop/share/hadoop/common/lib/xmlenc-0.52.jar:/home/guru99/Downloads/hadoop/share
/hadoop/common/lib/jsp-api-2.1.jar:/home/guru99/Downloads/hadoop/share/hadoop/co
mmon/lib/commons-collections-3.2.1.jar:/home/guru99/Downloads/hadoop/share/hadoo
p/common/lib/avro-1.7.4.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/li
b/jackson-core-asl-1.8.8.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/l
ib/commons-io-2.1.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/jers
ey-core-1.9.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/commons-co
dec-1.4.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/mockito-all-1.
8.5.jar:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/commons-cli-1.2.ja
r:/home/guru99/Downloads/hadoop/share/hadoop/common/lib/jets3t-0.6.1.jar:/home/g
uru99/Downloads/hadoop/share/hadoop/common/lib/guava-11.0.2.jar:/home/guru99/Dow
nloads/hadoop/share/hadoop/common/lib/commons-net-3.1.jar:/home/guru99/Downloads
/hadoop/share/hadoop/common/lib/commons-httpclient-3.1.jar:/home/guru99/Download
```

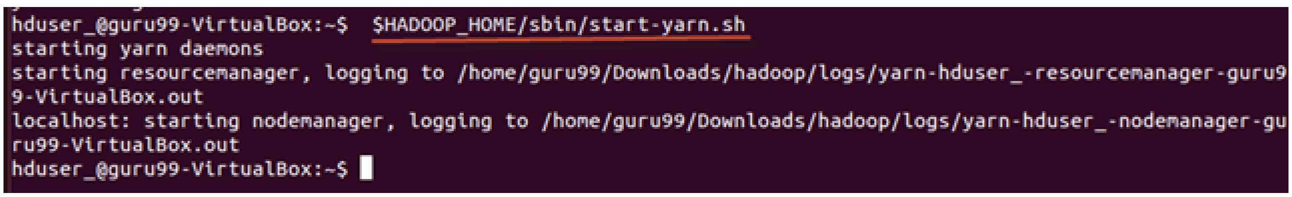
Step 5) Start Hadoop single node cluster using below command

```
$HADOOP_HOME/sbin/start-dfs.sh
```

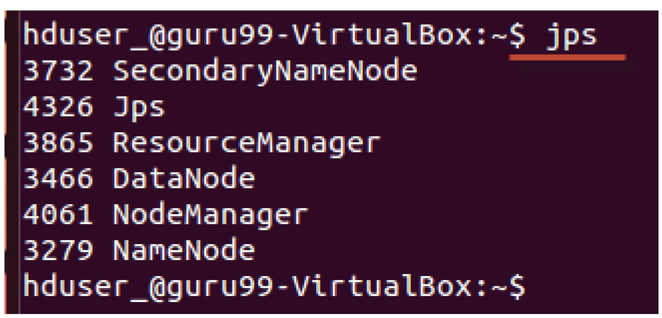
An output of above command




```
$HADOOP_HOME/sbin/start-yarn.sh
```



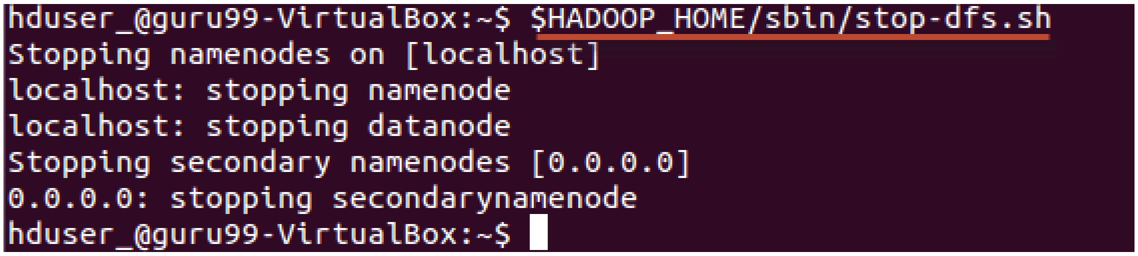
Using **'jps'** tool/command, verify whether all the Hadoop related processes are running or not.



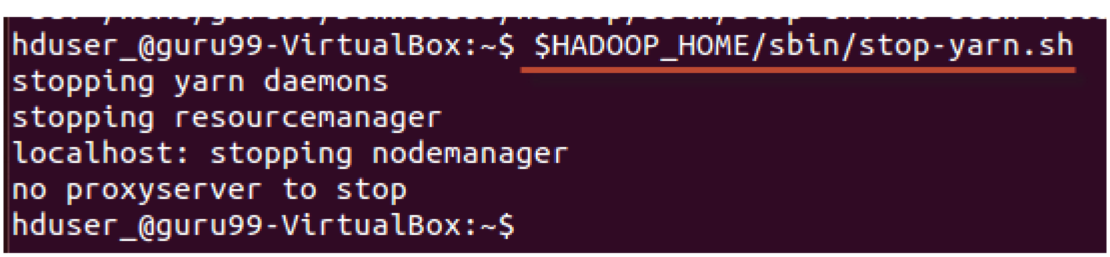
If Hadoop has started successfully then an output of jps should show NameNode, NodeManager, ResourceManager, SecondaryNameNode, DataNode.

Step 6) Stopping Hadoop

```
$HADOOP_HOME/sbin/stop-dfs.sh
```




```
$HADOOP_HOME/sbin/stop-yarn.sh
```



Report a Bug

Previous Prev

Next Continue

Daily Guru99 Newsletter

Kick off your day with the latest and most important AI news stories delivered right now.

Your Email:

you@emailaddress.com

Submit Now

Top Tutorials

About

About Us

Advertise with Us

Contact Us

Python

Testing

Hacking

Career Suggestion

SAP Career Suggestion Tool

Software Testing as a Career

Onpay

Omnisend

Xero

SAP

Java

SQL

Interesting

eBook

Selenium

Build Website

Linux

© Copyright - Guru99 2025